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Introductory Material

Introduction to Public Key Cryptography

GCD, Euclid's Algorithm and Bezout Coefficients

RSA Public Key Cryptosystem

Video: RSA Cryptography

43 min

Lab: Notes on RSA public-key cryptosystem

1h

Graded Assignment: Quiz on RSA

50 min

Basic Introduction to Quantum Computation

Problem Set: Week 1

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need a minimum score.

Quiz on RSA

Next item →

Review Learning Objectives

1 / 1 point

1. Calculate Euler's Totient Function $\varphi(33)$?

20

Assignment details

Correct

Due

May 5, 11:59 PM IST

Attempts

Unlimited

33 = 11 × 3 is the product of two prime numbers and thus $\varphi(33) = (11 - 1) \times (3 - 1) = 20$

Submitted

Apr 30, 6:27 PM IST

Try again

2. If p is a prime number, what is the value of $\varphi(p)$ the Euler Totient function of p ? We keep your highest score.

100%

1 / 1 point

View submission

See feedback

$p - 1$

Correct

For a prime number p , every number $1, \dots, p - 1$ is relatively prime to it.

Like

Dislike

Report an issue

3. What is the value of $\varphi(12)$? The Euler Totient function of the number 12?

1 / 1 point

4

Correct

The relatively prime numbers are 1, 5, 7, 11: since there are 4 of them, that yields the Euler Totient function

4. What is the value of $10^{60} \bmod 77$?

1 / 1 point

1

Correct

$60 = \varphi(77)$ and $10^{\varphi(77)} \bmod 77 = 1$

5. Let a be a number that is not divisible by 2 and not divisible by 5. What is the value of the last digit of the number a^a ?

1 / 1 point

☒ 1

☐ 2

☐ It depends on the choice of a

☐ 0

Correct

Clearly a is relatively prime with 10 and $4 = \varphi(10)$. Alternatively note that there are 4 such numbers 1, 3, 7, 9 and verify the answer for each of them.

6. Suppose we are designing RSA keys for $n = 11 \times 3 = 33$ and we choose $e = 7$. Complete the following calculations for the remaining keys.

6 / 6 points

Expression	Value
$\varphi(n)$	20 ✓ ☒
$e^{\varphi(n)} \bmod n$	1 ✓ ☒
$GCD(e, \varphi(n))$	1 ✓ ☒
(d, k) such that $d \times e - k \times \varphi(n) = 1$	(3, 1) ✓ ☒
Public Key	(7, 33) ✓ ☒
Private Key	(3, 33) ✓ ☒

Response 1

☐ 30

☒ 20

☐ 22

Correct

Response 2

☐ -1

☒ 1

☐ I need to use a calculator and find out since it can be anything.

Correct

Euler's theorem

Response 3

☐ 3

☐ 0

☒ 1

Correct

Response 4

☐ (1, 3)

☒ (3, 1)

☐ (6, 2)

Correct

Response 5

☒ (7, 33)

☐ (7, 20)

☐ (6, 20)

Correct

Response 6

☒ (3, 33)

☐ (1, 33)

☐ (2, 33)

Correct

7. RSA public keys are of the form (e, n) for two numbers e and n . Which of the following pieces of information, if known, will allow an attacker to compute the private key just from the public key?

1 / 1 point

☒ The prime factors of n .

Correct

☒ The Euler Totient function $\varphi(n)$

Correct

☐ The prime factors of e

☐ The value of $M^e \bmod n$ for some message M that is chosen by the attacker.